

1.
 1. The first main requirement for a symmetrical encryption algorithm is to have a strong encryption algorithm. Without this, the encoded message can easily be cracked. The secret key that both parties use must also be secured properly by both parties, and distributed in such a way that it is secure.
 2. The two basic functions in an encryption algorithm are Substitution, where elements of the plain text are replaced using some kind of rule, and Transportation, where the elements are rearranged.
 3. The Caesar Cipher is extremely vulnerable to brute force as there are only 25 possible rotations for a given string.
 4. One Time Pads (OTPs) have a few problems. The first is that a key must be at least as long as the message being encrypted. This leads to storage issues, as the data being encrypted effectively takes up twice as much space as it normally would. Another major issue is that each key is only usable once, requiring a secure way to distribute the key for every single message.
2.
 1. "homework", when shifted 4 letters to the right is "lsqiasvo".
 2. Encrypted: ISMFSL
 3. Encrypted: tietshssscemsaeiareg
3.
 1. Work on paper
 2. Work on Paper
 3. DES encryption does not provide authentication on its own, as there is no way in the algorithm to verify who sent the message. The message can be sent by anyone with the particular secret for the algorithm with no way to prove who sent it.
4. Answer Table:

	Number of bits
Plaintext block	64
Ciphertext block?	64
Key Length	64
Valid key length	56
Subkey Length	8
S-box input	48
S-box output	32
Input of a DES round	64 bits split in halves (32 and 32)
Input of a Mangler function	32

5. 5a and 5b are on the attached answer paper.
6. Work on paper.